

Kynema-FMB: A Performance-Portable Flexible-Multibody-Dynamics Solver

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Summary

Kynema-FMB (M. A. Sprague & Mudafort, 2022) is an open-source performance-portable flexible multibody (FMB) dynamics solver designed for time-domain simulations. Kynema-FMB was designed with a narrow focus, namely to provide a lightweight, fast, accurate FMB dynamics solver for coupling to computational-fluid-dynamics (CFD) codes, especially the broader suite of Kynema CFD codes (formerly called ExaWind) (Kuhn et al., 2025; Sharma et al., 2023; M. A. Sprague et al., 2020), for fluid-structure-interaction (FSI) simulations. Kynema-FMB is equipped to model systems that can be represented as a collection of beams and rigid bodies that are connected through constraints. Degrees of freedom are defined in the inertial/global frame of reference and include displacements and rotations (formally as rotation matrices, but stored as quaternions). The underlying formulation is built on a Lie-group time integrator designed for index-3 differential-algebraic equations that is second-order accurate in time (Brüls et al., 2012). Beam models are based on geometrically exact beam theory (Bauchau, 2011; Reissner, 1973) and are discretized as high-order spectral finite elements similar to those in BeamDyn (Wang et al., 2017). The governing equations for many FMB systems constitute a highly nonlinear system of constrained partial-differential equations. Kynema-FMB uses analytical Jacobians in the nonlinear-system solves in each time step. Linear systems use sparse storage and several third-party sparse-linear-system solvers are enabled. Ill conditioning of linear systems is mitigated with preconditioning described in Bottasso et al. (2008). Kynema-FMB is integrated with a simple open-source controller (Abbas et al., 2022). There is an application programming interface (API) for coupling to geometry-resolved CFD (like that in Sharma et al. (2023)) and actuator-force CFD (like that in Kuhn et al. (2025)). In the latter, for actuator-line models, Kynema-FMB includes an internal blade-element aerodynamics solver that depends on user-provided lookup tables for coefficients of lift and drag, i.e., aerodynamic polars. Kynema-FMB is written in C++ and leverages Kokkos and Kokkos-Kernels (Trott et al., 2021) as its performance portability layer enabling simulations on both CPU and GPU systems. The repository is equipped with extensive automated testing at the unit and regression/system levels.

Statement of need

The motivating use case for Kynema-FMB was to provide a modern structural-dynamics code for FSI coupling to the Kynema suite of open-source high-fidelity CFD solvers. Kynema-FMB fills the need for an open-source FMB structural dynamics code that is high fidelity, lightweight, robust, fast, and capable of running on different computer architectures, including GPUs from multiple manufacturers. Kynema-FMB is well suited for simulating structures that can be modeled as rigid bodies and slender deformable structures (i.e., beams), including offshore oil and gas infrastructure (floating platforms and mooring lines), power transmission lines, and rotor craft.

Kynema is written in C++ to facilitate integration with the C++-based Kynema CFD codes. While Kynema-FMB's primary target is coupling to the Kynema CFD suite, it is designed to be readily coupled with other fluid-dynamics codes as well. A driving need was also to have a code base that was developed using modern-software-development best practices to ensure the code is future proof and has minimal maintenance requirements. Best practices include test-driven-development, a multi-platform build system, nightly testing, thorough documentation, and extensive automated hierarchical testing.

We note that the Kynema CFD codes were first established with coupling to OpenFAST (J. M. Jonkman, 2013; "OpenFAST," n.d.), which includes models for structural dynamics, aerodynamics, hydrodynamics, control systems, drivetrain, etc. While OpenFAST was critical to the establishment and success of Kynema CFD codes, it faced challenges as a component in the Kynema suite due to its limited unit and regression testing and that it is Fortran code, which complicates creation and maintenance of APIs for coupling to C++ codes.

State of the field

There are several established open-source, C++-based FMB dynamics codes, including MBDyn (P. Masarati, 2014), Exudyn (Gerstmayr, 2023) (which is a follow on to HOTINT (Gerstmayr, 2009)), and Chrono (part of ProjectChrono) (Project Chrono Development Team, n.d.; Tasora et al., 2016). Kynema shares features with several of these codes, e.g., similar time-integration (ExuDyn and Chrono), use of the open-source Eign library for linear algebra (Chrono and HOTINT), and three-dimensional geometrically nonlinear beams (Chrono and MBDyn). Kynema-FMB stands apart in that it is:

- equipped with an extensive hierarchical automated test suite (currently 448 tests) that is run nightly and reported on CDash,
- built with Kokkos/Kokkos-Kernels enabling shared-memory CPU-based computing as well as GPU-based computing (on GPU hardware from multiple vendors),
- equipped with high-order spectral finite elements for beam discretization, and
- equipped for Spack-based (Gamblin et al., 2015) installation in addition to CMake.

Software design

Programming language and models

Kynema-FMB is written in C++ with tight integration of the Kokkos performance portability library. This approach allows a single code base to achieve near-optimal performance when run on CPU or any GPU platform, rather than requiring separate code paths. Optimized math routines, such as those covered by BLAS and LAPACK packages or sparse linear solvers, are obtained from specialized third-party libraries to ensure state-of-the-art performance.

Accessibility

Kynema-FMB provides a user-friendly, high level API for developers to use to define and run their structural dynamics problems and to couple Kynema-FMB with other codes. This approach decouples users of Kynema-FMB from the low level details of its implementation and improves the speed at which developers can define and execute their problem. Advanced users are able to use the lower level Kynema-FMB APIs directly or to define their own interfaces, should the high level APIs not address their needs directly.

Robustness

Kynema-FMB prioritizes software robustness through a comprehensive unit and regression test suite, which are run through a Continuous Integration process. Beyond that, Kynema-FMB continuously runs a variety of static and dynamic analysis tools to identify potential bugs. Linters and manual code review on every change help to ensure consistent, well designed, and sustainable software.

Performance

Kynema-FMB is performance-focused software. Core data structures are designed to provide optimal cache usage and all algorithms are written to best take advantage of on-chip resources. Kynema-FMB performs optimally on both CPU and GPU, using hierarchical parallelism and other techniques to ensure performance portability for problems of all sizes.

Research impact statement

As described in M. Sprague et al. (2025), Kynema-FMB has been

- successfully coupled to a blade-element aerodynamics solver and controller; simulations were compared against the OpenFAST simulations,
- successfully coupled to a two-phase Kynema FSI CFD simulation of a geometry resolved floating platform where Kynema-FMB solved the rigid body motion.

As described in the Kynema-FMB documentation ("Kynema-FMB," 2026), it has also been coupled to the Kynema-SGF structured-grid CFD code for actuator-line FSI simulations of the NREL 5-MW turbine (J. Jonkman et al., 2009) in a turbulent atmospheric boundary layer, with results compared against those simulated with OpenFAST. It was demonstrated that Kynema-FMB could provide stable, accurate solutions wherein the structure and CFD models used the same time step (Δt_{CFD}), whereas OpenFAST required a much smaller time step ($\Delta t_{\text{CFD}}/20$) for stability.

AI usage disclosure

Generative AI was not used in software creation, online documentaiton, or paper authoring.

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